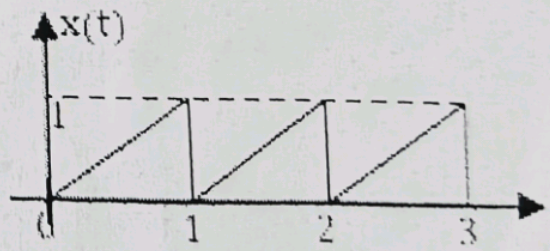
राष्ट्रीय प्रौद्योगिकी संस्थान, कुरुक्षेत्र

NATIONAL INSTITUTE OF TECHNOLOGY, KURUKSHETRA

DEPARTMENT OF ELECTRICAL ENGINEERING

Date: 25-09-2025 **Signals and Systems (RAPC204), Mid Term Exam-I** **Time: 50 Min.**

All questions are compulsory **Total: 15 Marks**

1. Each question carries equal marks [$(3 \times 2) = 6$ M]
   d) Consider a ramp signal $r(t) = t$ and sketch $x(t) = 3r(t-1) + r(-t+2)$ with a proper graphical representation?
   e) Check the periodicity of a signal $x(t) = 2\cos 5t + 1 - \sin 4t$. If periodic, determine the fundamental time Period of the signal?
   f) State the systems $y[n] + y[n-1] = x[n] + x[n-2]$ is linear, causal, time-invariant and stable?
2. Obtain the convolution of the signals $x(t) = e^{-3t}\{u(t) - u(t-2)\}$ and $h(t) = e^{-t}$ [5 M]
3. [4 M]

   Find the Fourier series of a signal shown in the figure?